## EXERCICE 1

| $X \backslash Y$ | $-2$ | $0$ | $1$ | $P_X$ |
|---|---|---|---|---|
| $-1$ | 0.15 | 0.15 | 0 | 0.3 |
| $0$ | 0.4 | 0 | 0 | 0.4 |
| $1$ | 0 | 0 | 0.3 | 0.3 |
| $P_Y$ | 0.55 | 0.15 | 0.3 | |

e.g. $P_{X,Y}(-1,0) = 0.15$

$P_Y(0) = 0.15$

$P_Y(1) = 0.3$

$P_X(0) = 0.4$

(c) $P(X \geq 0, Y \geq 0) = 0.3$

(d) $X, Y$ are independent $P_{X,Y}(x,y) = P_X(x)P_Y(y)$

$$\underbrace{P_X(0)}_{0.4}\underbrace{P_Y(0)}_{0.15} = 0.06 \neq 0 = P_{X,Y}(0,0)$$

not independent

EXERCISE 2

Consider the following two examples

$X$ : number of heads $\quad X \in \{0,1\}$

$Y$ : number of tails $\quad Y \in \{0,1\}$ ← FLIP ONE (FAIR) COIN

| $X \backslash Y$ | 0 | 1 | $P_X$ |
|---|---|---|---|
| 0 | 0 | 1/2 | 1/2 |
| 1 | 1/2 | 0 | 1/2 |
| $P_Y$ | 1/2 | 1/2 | |

MARGINALS:

$P_X(0) = P_X(1) = 1/2$

$P_Y(0) = P_Y(1) = 1/2$

$X$ : number of heads from coin 1 ← FLIP TWO (FAIR) COINS INDEPENDENTLY

$Y$ : number of heads from coin 2

| $X \backslash Y$ | 0 | 1 | $P_X$ |
|---|---|---|---|
| 0 | 1/4 | 1/4 | 1/2 |
| 1 | 1/4 | 1/4 | 1/2 |
| $P_Y$ | 1/2 | 1/2 | |

MARGINALS:

$P_X(0) = P_X(1) = 1/2$

$P_Y(0) = P_Y(1) = 1/2$

The marginals DO NOT determine the joint distribution in general. The example above is a counter example showing that two different joint distributions have the same marginals. However if $X, Y$ are independent, $P_X$ and $P_Y$ determine the joint distribution as $P_{X,Y}(x,y) = P_X(x) P_Y(y)$.

EXERCISE 3 $\quad x \in \{1,2,3,4,5,6\} \quad y \in \{0,1,2\}$

| $x \backslash y$ | 0 | 1 | 2 | $P_X$ |
|---|---|---|---|---|
| 1 | $1/12$ | $1/12$ | 0 | $1/6$ |
| 2 | $1/24$ | $1/12$ | $1/24$ | $1/6$ |
| 3 | $1/12$ | $1/12$ | 0 | $1/6$ |
| 4 | $1/24$ | $1/12$ | $1/24$ | $1/6$ |
| 5 | $1/12$ | $1/12$ | 0 | $1/6$ |
| 6 | $1/24$ | $1/12$ | $1/24$ | $1/6$ |
| $P_Y$ | $3/8$ | $1/2$ | $1/8$ | |

e.g. $P_{X,Y}(1,0) = P(X=1, Y=0) = \frac{1}{6} \cdot \frac{1}{2}$ intuitively as we need to score 1 with the die and flip a tail

e.g. $P_{X,Y}(2,0) = \frac{1}{6} \cdot \frac{1}{4} = 1/24$

c) $P_X(1) P_Y(2) = 1/6 \cdot 1/8 \neq 0 = P_{X,Y}(1,2)$

NOT INDEPENDENT!

EXERCISE 4

$$= 2 \cdot e^{-2x} \cdot e^{-y}$$

$$f_{X,Y}(x,y) = k \cdot e^{-2x-y} \qquad x,y \geq 0$$

(a) $$\int_0^{+\infty}\int_0^{+\infty} k \underbrace{e^{-2x-y}}_{e^{-2x}\,e^{-y}} \, dx\,dy = 1$$ as $f_{X,Y}$ is a joint pdf

$$= k \underbrace{\int_0^{+\infty} e^{-2x} \int_0^{+\infty} \boxed{e^{-y}} \, dy}_{=1} \, dx$$

pdf of Exp(1)

$= 1$ as it's the integral of an exponential pdf over its domain

otherwise $-e^{-y}\Big|_0^{+\infty} = 0 + e^{-0} = 1$

$$= k \int_0^{+\infty} e^{-2x} \, dx = 1$$

$$-\frac{1}{2}e^{-2\infty} + \frac{1}{2} \cdot e^{-0} = \frac{1}{2}$$

$$k \left[ e^{-2x} \cdot -\frac{1}{2} \right]_0^{+\infty} = k \, \frac{1}{2} \cdot 1 = 1 \Rightarrow k = 2$$

(b) $$f_X(x) = \int_0^{+\infty} 2e^{-2x} e^{-y} \, dy = 2e^{-2x} \int_0^{+\infty} e^{-y} \, dy = 2e^{-2x}$$

$\Rightarrow X \sim Exp(2)$

$$f_Y(y) = \int_0^{+\infty} 2e^{-2x} e^{-y} \, dx = e^{-y} \qquad Y \sim Exp(1)$$

$X \sim Exp(2)$

$Y \sim Exp(1)$

c) Are $X, Y$ independent?

$$P_X(x)P_Y(y) = 2e^{-2x}\, e^{-y} = 2e^{-2x-y} = P_{X,Y}(x,y)$$

product of marginals

YESSSS!

joint pdf

d) $P(X \geq 1, Y \geq 1) = P(X \geq 1)\, P(Y \geq 1)$

INDIPENDENCE

* with R Studio

$$= \Big(1 - \underbrace{\text{pexp}(1, \overset{\lambda}{2})}_{P(X \leq 1)}\Big)\Big(1 - \text{pexp}(1, 1)\Big) = 0.04979$$

* by HAND

$$= \underbrace{\int_1^{+\infty} 2e^{-2x}\,dx}_{-e^{-2x}\Big|_1^{+\infty}} \underbrace{\int_1^{+\infty} e^{-y}\,dy}_{-e^{-y}\Big|_1^{+\infty}} = e^{-2} \cdot e^{-1} = e^{-3}$$